Interpreting Weights and Persistence of Pre-training Traits

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The Two-Part Question

A student asks:

1. **Can inspecting weights reveal a model's tendencies?**
2. **Do models revert to pre-training traits after deployment?**

Both touch the central interpretability goal: connecting the *static artifact* (parameters) to *behavioral dispositions*.

Key caveat up front: weights are necessary but *not transparent*. Function is distributed, polysemantic, and context-dependent [1, 2].

What Weight Inspection Actually Buys You

Direct reading of raw parameter matrices is rarely informative. Productive methods instead *structure* the weights:

- **Circuit analysis**: trace how attention heads and MLPs compose to implement algorithms [3, 4].
- **Feature decomposition**: sparse autoencoders recover interpretable directions from superposed activations [5, 6].
- **Weight-space probes**: logit lens / causal tracing localize where factual associations live [7, 8].

Tendencies become visible as *reusable mechanisms*, not as single numbers.

From Mechanism to "Tendency"

A behavioral tendency is an emergent regularity. Interpretability links it to weights by:

- **Causal intervention**: ablate or patch a component and measure the behavioral delta [4, 7].
- **Feature attribution**: identify features whose activation predicts a disposition (e.g., sycophancy, refusal).
- **Steering vectors**: adding/subtracting activation directions shifts outputs, evidence that the direction encodes the trait [9, 10].

Caution: correlation between a direction and behavior is not proof of a monolithic "trait neuron"; features are superposed and polysemantic [1].

The Reversion Question: What "Deployment" Changes

Frozen deployment weights *do not drift*: inference does not update parameters. So "reversion" is not literal weight decay. What varies is:

- **Context**: prompts and few-shot examples can surface latent pre-training behaviors suppressed by fine-tuning.
- **Distribution shift**: inputs unlike alignment data fall back toward base-model priors.
- **Jailbreaks**: adversarial prompts route around safety fine-tuning [11, 12].

Fine-tuning often adds a thin behavioral layer over durable pre-training knowledge.

Evidence That Pre-training Traits Persist

- **Superficial alignment**: instruction tuning adjusts style/format while most capability comes from pre-training [13].
- **Shallow safety**: safety behavior concentrated in the first few tokens is easy to bypass, implying underlying tendencies remain [14].
- **Fine-tuning fragility**: a few benign steps can remove safety alignment, exposing base dispositions [15].
- **Sleeper agents**: backdoored behaviors survive standard safety training [16].

Together: pre-training traits are latent and *recoverable*, not erased.

Synthesis: Answering the Student

Q1. Weight inspection, via circuits, feature decomposition, and causal intervention, can reveal *mechanisms* that underlie tendencies, with the caveats of superposition and the gap between mechanism and behavior.

Q2. Deployed weights are static, so there is no spontaneous reversion. But because alignment is often shallow, pre-training tendencies can *re-emerge* under distribution shift, adversarial prompts, or further fine-tuning.

Practical stance: treat alignment as a thin, bypassable layer; audit for latent dispositions using interpretability plus red-teaming.

Takeaways and Open Problems

- Tendencies are *distributed*; no single weight "is" a trait.
- Causal methods beat passive weight-reading for attributing behavior.
- Pre-training priors persist beneath fine-tuning and can resurface.
- **Open**: scaling SAEs, verifying feature causality, and measuring how "deep" alignment truly goes [5, 14].

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